Measuring a Response to Intervention Model in Early Childhood: Examining Assessments for Identification, Decision Making and Progress Monitoring

Alisha Wackerle-Hollman, Tracy Bradfield, Scott McConnell & Trina D. Spencer

Center for Early Education and Development: U of MN, The Ohio State University &
The Center for Response to Intervention in Early Childhood

Individual Growth and Development Indicators: Assessment Systems- Screening, Progress Monitoring and Decision Making Process

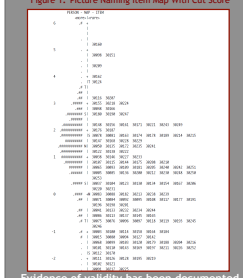
Progress Monitoring within a Tier 2 Vocabulary and Comprehension Intervention: Assessment of Story Comprehension (ASC)

As response to intervention (RTI) is increasingly adopted within early childhood education programs, the need for robust assessments suited for use to identify/screen, evaluate and progress monitor student performance has become more and more salient. Psychometrically sound and logistically efficient measures will provide a basis to effectively identify preschool age students in need of intervention, to evaluate the level of intervention warranted (e.g. Tier 2 or Tier 3) and to monitor progress while receiving suggested level of intervention.

Screening/ Identification Assessment (ID-IGDIs)

The Individual Growth and Development Indicators (IGDIs) are early literacy measures specifically designed for use within an RTI model. ID-IGDIs 2.0 were specifically designed to determine if a student's score on the measures is characteristic of Tier 2 or Tier 3 level performance. The measures were designed using IRT/Rasch analysis such that each item has a difficulty level associated with it. ID-IGDIs 2.0 are the first of their kind to produce an ability score for students in reference to an identified cut score between Tier 1 performance levels (performing adequately in the universal curriculum) and Tier 2/3 performance level (in need of supplemental intervention to be successful).

Figure 1. Picture Naming Item Map With Cut Score



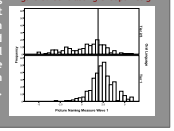
Evidence of validity has been documented through criterion correlations with accepted measures of early literacy (*r = . 45-.80*). Reliability statistic alpha yielded . 93-.95 across the 5 IGDI measures.

Figure 2. ID-IGDI 2.0 Screening Measures

5 IGDI measures pictured in Figure 2 (with the exception of First Sounds) are designed to capture seasonal performance across four early literacy domains: oral language, phonological awareness, alphabet knowledge and comprehension. The measures are designed to be 15 items long and are given in a specific sequence. Screening items are selected to represent performance at the cut-score location.

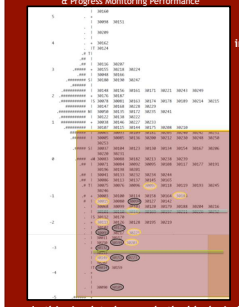
Seasonal cut scores are achieved using contrasting groups design such that student performance on IGDIs and criterion referenced assessments were contrasted with teacher ratings on performance level descriptors. Sensitivity and Specificity were evaluated using ROC analysis, with sensitivity set at .70.

Figure 3. Contrasting Groups Design



Progress Monitoring Assessment (IGDI-PM)

Figure 4. Picture Naming Item Map With Cut Score & Progress Monitoring Performance

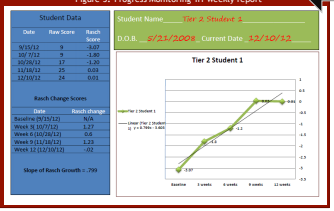


In this simulated example the black circled items represent correct performance at Time 1 with a computed Rasch ability score at the green line. The yellow circled items represent correct performance at Tier 2 with a computed Rasch ability score at the blue line. This change is ability is detectable using a Rasch score, but would not be using a raw score as in both Times the raw score is 7.

For those students selected to receive Tier 2 or Tier 3 intervention the IGDI progress monitoring set can be used to evaluate growth over time. The progress monitoring sets (2 parallel sets) are designed to include 30 items given at 3 week frequencies for each of 5 measures. Assessors or interventionists should only use the IGDI measures that match the content of the intervention being delivered (e.g. if an intervention in vocabulary is given , the Picture Naming IGDI should be used for progress monitoring).

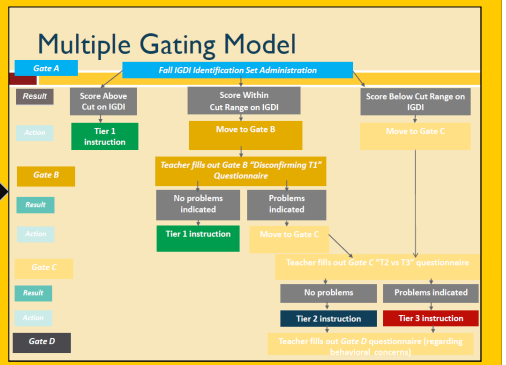
Items used for progress monitoring are specifically selected from item locations (difficulties) representative of Tier 2 and Tier 3 performance (shaded in red in Figure 4). For each assessment Rasch ability scores are computed (rather than raw scores) to examine how student performance changes over brief periods of time based on the exact items the child can answer correctly. An example of tri-weekly output is provided in Figure 5.

Figure 5. Progress Monitoring Tri-weekly report



Decision Making Framework (IGDI-DMF)

Cut scores that accompany the ID-IGDIs, distinguish between children demonstrating adequate levels of performance given the general classroom instruction (Tier One candidates) and those who would likely benefit from more intensive levels of instruction (Tier Two or Three). To further distinguish between those who would benefit from Tier two versus Tier Three instruction, a multiple-gating decision making framework has been developed and is currently being evaluated.



This DMF starts with child score on IGDI as the first gate. Teachers are then asked to complete a teacher questionnaire that solicits additional information about child ability in the given domains to inform decision making at the successive gates.

CRTIEC Decision Making Framework Teacher Questionnaire: Oral Language & Comprehension

DIRECTIONS:

For each question, make a check mark in the box that best indicates how often each child displays each skill or behavior. Please check **only ONE box** for each question.

Part I. Language and Comprehension Skills	Rarely or Never	Less Often	More Often	Frequently Or Always
1. This child uses grammatically correct sentences when speaking.				
2. This child learns new words without direct instruction.				
3. This child uses adjectives to describe objects, places or things. (Ex: I want the RED ball).				
4. This child tells simple stories that make sense.				
5. This child can answer questions about events that happened in the past.				
6. This child can tell you what they did over the weekend.				
7. This child asks Who, What, Where, When, How questions.				
8. When asked, this child can answer: Who, What, Where, When, How questions.				
9. When this child hears a word they don't know, they ask what it means.				
10. When asked questions about a story, this child can answer correctly.				
11. This child understands simple stories.				
12. This child can retell simple stories.				
13. When listening to a simple story, this child can make predictions about what will happen next.				

Validation of the DMF model is occurring this academic year (2012-2013) in over 100 classrooms across 3 states serving Tier 1, Tier 2 and Tier 3 students. Preliminary results suggest the DMF model indicates high levels of reliability at alphas of range from .961 to .987

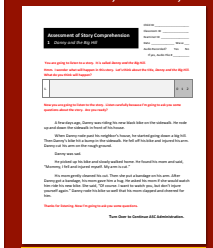
Evidence to support empirical standards of validity will be collected this year using the CELF-Preschool 2 for Oral language in fall and the TOPEL for Phonological Awareness in winter and spring.

Through analysis of children's score on the IGDI measures, in combination with the results of the teacher questionnaire, tier placement recommendations are made.

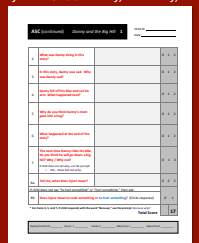
Student ID	Construct Domain to Target	IGDI Task	IGDI Score	TQ Result	Tier Assignment
1105001	Oral Language	PN	1	Did Not Pass	Tier 3
	Comprehension	WDDB	4		
1105002	Oral Language	PN	8	Pass	Tier 1
	Comprehension	WDDB	10		
110503	Oral Language	PN	4	Pass	Tier 2
	Comprehension	WDDB	13		
110504	Oral Language	PN	11	N/A	Tier 1
	Comprehension	WDDB	14		

Assessment of Story Comprehension (ASC)

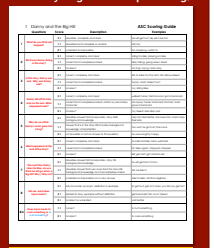
The ASC was developed to measure treatment effects of a tier 2 vocabulary and comprehension intervention called Story Friends. It can also be used to monitor language development across the preschool year. The creation and validation of the ASC has followed an iterative process. As a curriculum-based measure, the ASC samples authentic child behaviors that reflect key outcomes, has standardized administration and scoring procedures, and is time efficient, economical, and easy to use. ASC validity, reliability, and sensitivity to growth are promising.



Children listen to a brief relatable story (e.g., breaking a toy, spilling paint, asking someone to play).



Examiner asks a mix of recall and inferential questions, which are consistent across forms.



Answers are recorded in real time. Then scored using a story specific scoring guide.

All 9 stories have the same story structure, linguistic complexity, length, and contextual support for a rare word.

One question asks for the definition of the rare word. If child cannot give definition a choice of 2 definitions is offered.

Administration takes about 3 minutes for 1 story (progress monitoring) or 8-10 minutes for 3 stories in a session (benchmark).

ASC Iterative Development

DEVELOPMENT

Created research-based framework. Wrote 9 stories and questions. Developed a single scoring guide.

STUDY 1

36 preschoolers received 3 of the 9 ASC forms and criterion measures. Preliminary evidence of validity, inter-scorer reliability, and implementation of fidelity.

REVISIONS

Rewrote 3 stories, eliminated 1 question, and developed story specific scoring guides.

STUDY 2

20 preschoolers received all 9 forms and criterion measures. Additional evidence of validity, inter-scorer reliability, and implementation fidelity.

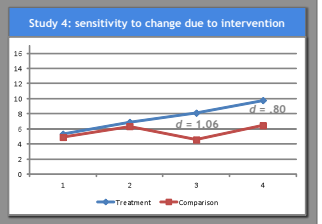
STUDY 3

76 preschoolers received all 9 forms and criterion measures. Validity coefficient with CELF SS is medium (.38). Fidelity of implementation is very high (97-100%). Scoring reliability is adequate (89%). Alternate form reliability is moderate (mean = .72)

STUDY 4

18 preschoolers. The treatment group participated in the tier 2 vocabulary and comprehension intervention. ASC was administered pre-intervention, post-unit 1, post-unit 2, and post unit 3.

Study 4: sensitivity to change due to intervention



University of Minnesota Driven to Discover℠

Center for Early Education and Development
Urban Research and Outreach-Engagement Center
2001 Plymouth Avenue N., Minneapolis, MN 55411

Contacts: Alisha Wackerle-Hollman: wacke020@umn.edu
Tracy Bradfield: tab@umn.edu
Trina Spencer: trina.spencer@nau.edu

ies INSTITUTE OF EDUCATION SCIENCES